

## nsys profile for [v1.cu](#) (4096)

[5/8] Executing 'cuda\_api\_sum' stats report

| Time (%) | Total Time (ns) | Num Calls | Avg (ns)      | Med (ns)      | Min (ns)    | Max (ns)    | StdDev (ns) | Name                   |
|----------|-----------------|-----------|---------------|---------------|-------------|-------------|-------------|------------------------|
| 93.1     | 172,669,150     | 1         | 172,669,150.0 | 172,669,150.0 | 172,669,150 | 172,669,150 | 0.0         | cudaMallocManaged      |
| 5.0      | 9,247,063       | 10        | 924,706.3     | 906,747.0     | 875,892     | 1,022,880   | 42,678.7    | cudaDeviceSynchronize  |
| 1.5      | 2,824,732       | 1         | 2,824,732.0   | 2,824,732.0   | 2,824,732   | 2,824,732   | 0.0         | cuLibraryLoadData      |
| 0.3      | 505,480         | 10        | 50,548.0      | 6,274.5       | 5,178       | 367,958     | 112,860.5   | cudaLaunchKernel       |
| 0.2      | 311,972         | 1         | 311,972.0     | 311,972.0     | 311,972     | 311,972     | 0.0         | cudaFree               |
| 0.0      | 4,973           | 10        | 497.3         | 227.0         | 154         | 1,685       | 565.9       | cuKernelGetName        |
| 0.0      | 1,206           | 1         | 1,206.0       | 1,206.0       | 1,206       | 1,206       | 0.0         | cuModuleGetLoadingMode |
| 0.0      | 779             | 1         | 779.0         | 779.0         | 779         | 779         | 0.0         | cuLibraryGetKernel     |

[6/8] Executing 'cuda\_gpu\_kern\_sum' stats report

| Time (%) | Total Time (ns) | Instances | Avg (ns)  | Med (ns)  | Min (ns) | Max (ns) | StdDev (ns) | Name                          |
|----------|-----------------|-----------|-----------|-----------|----------|----------|-------------|-------------------------------|
| 100.0    | 7,299,298       | 10        | 729,929.8 | 696,214.5 | 688,838  | 876,680  | 63,007.6    | bodyForce(Body *, float, int) |

[7/8] Executing 'cuda\_gpu\_mem\_time\_sum' stats report

| Time (%) | Total Time (ns) | Count | Avg (ns) | Med (ns) | Min (ns) | Max (ns) | StdDev (ns) | Operation                            |
|----------|-----------------|-------|----------|----------|----------|----------|-------------|--------------------------------------|
| 56.2     | 184,889         | 18    | 10,271.6 | 9,983.5  | 4,416    | 16,639   | 5,994.7     | [CUDA memcpy Unified Host-to-Device] |
| 43.8     | 143,853         | 40    | 3,596.3  | 3,519.5  | 1,407    | 10,560   | 2,328.5     | [CUDA memcpy Unified Device-to-Host] |

[8/8] Executing 'cuda\_gpu\_mem\_size\_sum' stats report

| Total (MB) | Count | Avg (MB) | Med (MB) | Min (MB) | Max (MB) | StdDev (MB) | Operation                            |
|------------|-------|----------|----------|----------|----------|-------------|--------------------------------------|
| 1.311      | 40    | 0.033    | 0.033    | 0.004    | 0.061    | 0.029       | [CUDA memcpy Unified Device-to-Host] |
| 1.180      | 18    | 0.066    | 0.066    | 0.033    | 0.098    | 0.034       | [CUDA memcpy Unified Host-to-Device] |

average execution time for [v1\\_2.cu](#) (65536)

0.3968 sec

## nsys profile for [v2.cu](#) (4096)

[5/8] Executing 'cuda\_api\_sum' stats report

| Time (%) | Total Time (ns) | Num Calls | Avg (ns)      | Med (ns)      | Min (ns)    | Max (ns)    | StdDev (ns) | Name                   |
|----------|-----------------|-----------|---------------|---------------|-------------|-------------|-------------|------------------------|
| 92.0     | 161,020,736     | 1         | 161,020,736.0 | 161,020,736.0 | 161,020,736 | 161,020,736 | 0.0         | cudaMallocManaged      |
| 7.3      | 12,697,786      | 10        | 1,269,778.6   | 1,266,815.0   | 1,180,337   | 1,340,466   | 45,701.5    | cudaDeviceSynchronize  |
| 0.4      | 641,996         | 10        | 64,199.6      | 28,920.0      | 5,143       | 443,486     | 134,130.0   | cudaLaunchKernel       |
| 0.2      | 422,594         | 1         | 422,594.0     | 422,594.0     | 422,594     | 422,594     | 0.0         | cuLibraryLoadData      |
| 0.1      | 132,988         | 1         | 132,988.0     | 132,988.0     | 132,988     | 132,988     | 0.0         | cudaFree               |
| 0.0      | 10,154          | 10        | 1,015.4       | 1,338.0       | 161         | 2,153       | 745.4       | cuKernelGetName        |
| 0.0      | 1,108           | 1         | 1,108.0       | 1,108.0       | 1,108       | 1,108       | 0.0         | cuModuleGetLoadingMode |
| 0.0      | 666             | 1         | 666.0         | 666.0         | 666         | 666         | 0.0         | cuLibraryGetKernel     |

[6/8] Executing 'cuda\_gpu\_kern\_sum' stats report

| Time (%) | Total Time (ns) | Instances | Avg (ns)    | Med (ns)    | Min (ns)  | Max (ns)  | StdDev (ns) | Name                          |
|----------|-----------------|-----------|-------------|-------------|-----------|-----------|-------------|-------------------------------|
| 100.0    | 10,790,182      | 10        | 1,079,018.2 | 1,061,175.0 | 1,033,558 | 1,141,912 | 39,729.7    | bodyForce(Body *, float, int) |

[7/8] Executing 'cuda\_gpu\_mem\_time\_sum' stats report

| Time (%) | Total Time (ns) | Count | Avg (ns) | Med (ns) | Min (ns) | Max (ns) | StdDev (ns) | Operation                            |
|----------|-----------------|-------|----------|----------|----------|----------|-------------|--------------------------------------|
| 68.7     | 305,686         | 90    | 3,396.5  | 3,471.5  | 2,111    | 4,543    | 822.6       | [CUDA memcpy Unified Host-to-Device] |
| 31.3     | 139,376         | 40    | 3,484.4  | 3,535.5  | 1,407    | 5,856    | 2,055.0     | [CUDA memcpy Unified Device-to-Host] |

[8/8] Executing 'cuda\_gpu\_mem\_size\_sum' stats report

| Total (MB) | Count | Avg (MB) | Med (MB) | Min (MB) | Max (MB) | StdDev (MB) | Operation                            |
|------------|-------|----------|----------|----------|----------|-------------|--------------------------------------|
| 1.311      | 40    | 0.033    | 0.033    | 0.004    | 0.061    | 0.029       | [CUDA memcpy Unified Device-to-Host] |
| 1.180      | 90    | 0.013    | 0.012    | 0.004    | 0.033    | 0.008       | [CUDA memcpy Unified Host-to-Device] |

average execution time for [v2\\_2.cu](#) (65536)

0.3849 sec

## nsys profile for [v3.cu](#) (4096)

[5/8] Executing 'cuda\_api\_sum' stats report

| Time (%) | Total Time (ns) | Num Calls | Avg (ns)      | Med (ns)      | Min (ns)    | Max (ns)    | StdDev (ns)  | Name                   |
|----------|-----------------|-----------|---------------|---------------|-------------|-------------|--------------|------------------------|
| 97.3     | 5,848,910,989   | 10        | 584,891,098.9 | 579,654,768.0 | 579,446,313 | 631,902,382 | 16,518,769.2 | cudaDeviceSynchronize  |
| 2.6      | 155,748,851     | 1         | 155,748,851.0 | 155,748,851.0 | 155,748,851 | 155,748,851 | 0.0          | cudaMallocManaged      |
| 0.0      | 2,831,431       | 1         | 2,831,431.0   | 2,831,431.0   | 2,831,431   | 2,831,431   | 0.0          | cuLibraryLoadData      |
| 0.0      | 654,210         | 10        | 65,421.0      | 31,006.0      | 13,388      | 377,624     | 110,175.1    | cudaLaunchKernel       |
| 0.0      | 190,320         | 1         | 190,320.0     | 190,320.0     | 190,320     | 190,320     | 0.0          | cudaFree               |
| 0.0      | 11,848          | 10        | 1,184.8       | 1,294.0       | 163         | 1,966       | 564.9        | cuKernelGetName        |
| 0.0      | 1,160           | 1         | 1,160.0       | 1,160.0       | 1,160       | 1,160       | 0.0          | cuModuleGetLoadingMode |
| 0.0      | 773             | 1         | 773.0         | 773.0         | 773         | 773         | 0.0          | cuLibraryGetKernel     |

[6/8] Executing 'cuda\_gpu\_kern\_sum' stats report

| Time (%) | Total Time (ns) | Instances | Avg (ns)      | Med (ns)      | Min (ns)    | Max (ns)    | StdDev (ns)  | Name                          |
|----------|-----------------|-----------|---------------|---------------|-------------|-------------|--------------|-------------------------------|
| 100.0    | 5,848,668,425   | 10        | 584,866,842.5 | 579,650,238.0 | 579,444,880 | 631,891,029 | 16,523,134.2 | bodyForce(Body *, float, int) |

[7/8] Executing 'cuda\_gpu\_mem\_time\_sum' stats report

| Time (%) | Total Time (ns) | Count | Avg (ns) | Med (ns) | Min (ns) | Max (ns) | StdDev (ns) | Operation                            |
|----------|-----------------|-------|----------|----------|----------|----------|-------------|--------------------------------------|
| 62.0     | 211,828         | 36    | 5,884.1  | 5,039.5  | 1,951    | 15,584   | 4,091.1     | [CUDA memcpy Unified Host-to-Device] |
| 38.0     | 129,777         | 40    | 3,244.4  | 3,216.0  | 1,119    | 5,536    | 2,136.8     | [CUDA memcpy Unified Device-to-Host] |

[8/8] Executing 'cuda\_gpu\_mem\_size\_sum' stats report

| Total (MB) | Count | Avg (MB) | Med (MB) | Min (MB) | Max (MB) | StdDev (MB) | Operation                            |
|------------|-------|----------|----------|----------|----------|-------------|--------------------------------------|
| 1.311      | 40    | 0.033    | 0.033    | 0.004    | 0.061    | 0.029       | [CUDA memcpy Unified Device-to-Host] |
| 1.180      | 36    | 0.033    | 0.033    | 0.004    | 0.061    | 0.029       | [CUDA memcpy Unified Host-to-Device] |

average execution time for [v3\\_2.cu](#) (65536)

3163.9620 sec

## nsys profile for [v4.cu](#) (4096)

[5/8] Executing 'cuda\_api\_sum' stats report

| Time (%) | Total Time (ns) | Num Calls | Avg (ns)      | Med (ns)      | Min (ns)    | Max (ns)    | StdDev (ns) | Name                   |
|----------|-----------------|-----------|---------------|---------------|-------------|-------------|-------------|------------------------|
| 88.3     | 159,905,573     | 1         | 159,905,573.0 | 159,905,573.0 | 159,905,573 | 159,905,573 | 0.0         | cudaMallocManaged      |
| 11.0     | 19,928,570      | 10        | 1,992,857.0   | 1,992,413.5   | 1,950,513   | 2,054,545   | 34,287.2    | cudaDeviceSynchronize  |
| 0.4      | 641,781         | 10        | 64,178.1      | 31,000.0      | 5,805       | 383,133     | 112,717.8   | cudaLaunchKernel       |
| 0.2      | 446,911         | 1         | 446,911.0     | 446,911.0     | 446,911     | 446,911     | 0.0         | cuLibraryLoadData      |
| 0.1      | 227,703         | 1         | 227,703.0     | 227,703.0     | 227,703     | 227,703     | 0.0         | cudaFree               |
| 0.0      | 11,422          | 10        | 1,142.2       | 1,386.5       | 156         | 1,544       | 534.5       | cuKernelGetName        |
| 0.0      | 1,011           | 1         | 1,011.0       | 1,011.0       | 1,011       | 1,011       | 0.0         | cuModuleGetLoadingMode |
| 0.0      | 608             | 1         | 608.0         | 608.0         | 608         | 608         | 0.0         | cuLibraryGetKernel     |

[6/8] Executing 'cuda\_gpu\_kern\_sum' stats report

| Time (%) | Total Time (ns) | Instances | Avg (ns)    | Med (ns)    | Min (ns)  | Max (ns)  | StdDev (ns) | Name                          |
|----------|-----------------|-----------|-------------|-------------|-----------|-----------|-------------|-------------------------------|
| 100.0    | 17,996,567      | 10        | 1,799,656.7 | 1,771,723.0 | 1,741,610 | 2,056,402 | 94,154.2    | bodyForce(Body *, float, int) |

[7/8] Executing 'cuda\_gpu\_mem\_time\_sum' stats report

| Time (%) | Total Time (ns) | Count | Avg (ns) | Med (ns) | Min (ns) | Max (ns) | StdDev (ns) | Operation                            |
|----------|-----------------|-------|----------|----------|----------|----------|-------------|--------------------------------------|
| 60.3     | 212,627         | 36    | 5,906.3  | 6,335.5  | 2,207    | 12,224   | 2,610.2     | [CUDA memcpy Unified Host-to-Device] |
| 39.7     | 139,917         | 40    | 3,497.9  | 3,519.5  | 1,407    | 5,888    | 2,083.2     | [CUDA memcpy Unified Device-to-Host] |

[8/8] Executing 'cuda\_gpu\_mem\_size\_sum' stats report

| Total (MB) | Count | Avg (MB) | Med (MB) | Min (MB) | Max (MB) | StdDev (MB) | Operation                            |
|------------|-------|----------|----------|----------|----------|-------------|--------------------------------------|
| 1.311      | 40    | 0.033    | 0.033    | 0.004    | 0.061    | 0.029       | [CUDA memcpy Unified Device-to-Host] |
| 1.180      | 36    | 0.033    | 0.033    | 0.004    | 0.061    | 0.021       | [CUDA memcpy Unified Host-to-Device] |

average execution time for [v4\\_2.cu](#) (65536)

3.5686 sec

## nsys profile for [v5.cu](#) (4096)

[5/8] Executing 'cuda\_api\_sum' stats report

| Time (%) | Total Time (ns) | Num Calls | Avg (ns)      | Med (ns)      | Min (ns)    | Max (ns)    | StdDev (ns) | Name                   |
|----------|-----------------|-----------|---------------|---------------|-------------|-------------|-------------|------------------------|
| 91.3     | 153,957,968     | 1         | 153,957,968.0 | 153,957,968.0 | 153,957,968 | 153,957,968 | 0.0         | cudaMallocManaged      |
| 7.7      | 13,015,699      | 10        | 1,301,569.9   | 1,294,914.5   | 1,232,147   | 1,363,718   | 37,723.4    | cudaDeviceSynchronize  |
| 0.5      | 847,795         | 10        | 84,779.5      | 37,645.5      | 28,345      | 446,845     | 129,028.6   | cudaLaunchKernel       |
| 0.4      | 626,069         | 1         | 626,069.0     | 626,069.0     | 626,069     | 626,069     | 0.0         | cuLibraryLoadData      |
| 0.1      | 241,708         | 1         | 241,708.0     | 241,708.0     | 241,708     | 241,708     | 0.0         | cudaFree               |
| 0.0      | 16,089          | 10        | 1,608.9       | 1,545.5       | 1,034       | 2,137       | 338.0       | cuKernelGetName        |
| 0.0      | 1,175           | 1         | 1,175.0       | 1,175.0       | 1,175       | 1,175       | 0.0         | cuModuleGetLoadingMode |
| 0.0      | 663             | 1         | 663.0         | 663.0         | 663         | 663         | 0.0         | cuLibraryGetKernel     |

[6/8] Executing 'cuda\_gpu\_kern\_sum' stats report

| Time (%) | Total Time (ns) | Instances | Avg (ns)    | Med (ns)    | Min (ns)  | Max (ns)  | StdDev (ns) | Name                          |
|----------|-----------------|-----------|-------------|-------------|-----------|-----------|-------------|-------------------------------|
| 100.0    | 11,123,423      | 10        | 1,112,342.3 | 1,081,395.0 | 1,026,627 | 1,367,236 | 95,203.6    | bodyForce(Body *, float, int) |

[7/8] Executing 'cuda\_gpu\_mem\_time\_sum' stats report

| Time (%) | Total Time (ns) | Count | Avg (ns) | Med (ns) | Min (ns) | Max (ns) | StdDev (ns) | Operation                            |
|----------|-----------------|-------|----------|----------|----------|----------|-------------|--------------------------------------|
| 64.3     | 300,689         | 90    | 3,341.0  | 3,359.0  | 2,143    | 4,992    | 834.1       | [CUDA memcpy Unified Host-to-Device] |
| 35.7     | 166,796         | 40    | 4,169.9  | 4,175.5  | 1,407    | 16,480   | 3,480.4     | [CUDA memcpy Unified Device-to-Host] |

[8/8] Executing 'cuda\_gpu\_mem\_size\_sum' stats report

| Total (MB) | Count | Avg (MB) | Med (MB) | Min (MB) | Max (MB) | StdDev (MB) | Operation                            |
|------------|-------|----------|----------|----------|----------|-------------|--------------------------------------|
| 1.311      | 40    | 0.033    | 0.033    | 0.004    | 0.061    | 0.029       | [CUDA memcpy Unified Device-to-Host] |
| 1.180      | 90    | 0.013    | 0.012    | 0.004    | 0.033    | 0.008       | [CUDA memcpy Unified Host-to-Device] |

## average execution time for v5\_2.cu (65536)

1.3889 sec

## nsys profile for [v6.cu](#) (4096)

[5/8] Executing 'cuda\_api\_sum' stats report

| Time (%) | Total Time (ns) | Num Calls | Avg (ns)      | Med (ns)      | Min (ns)    | Max (ns)    | StdDev (ns) | Name                   |
|----------|-----------------|-----------|---------------|---------------|-------------|-------------|-------------|------------------------|
| 91.4     | 164,740,591     | 1         | 164,740,591.0 | 164,740,591.0 | 164,740,591 | 164,740,591 | 0.0         | cudaMallocManaged      |
| 8.0      | 14,342,300      | 10        | 1,434,230.0   | 1,435,853.0   | 1,357,325   | 1,527,153   | 58,987.2    | cudaDeviceSynchronize  |
| 0.3      | 537,215         | 10        | 53,721.5      | 17,934.5      | 5,190       | 382,303     | 116,156.9   | cudaLaunchKernel       |
| 0.3      | 451,894         | 1         | 451,894.0     | 451,894.0     | 451,894     | 451,894     | 0.0         | cuLibraryLoadData      |
| 0.1      | 147,091         | 1         | 147,091.0     | 147,091.0     | 147,091     | 147,091     | 0.0         | cudaFree               |
| 0.0      | 7,594           | 10        | 759.4         | 562.5         | 157         | 1,619       | 621.2       | cuKernelGetName        |
| 0.0      | 1,478           | 1         | 1,478.0       | 1,478.0       | 1,478       | 1,478       | 0.0         | cuModuleGetLoadingMode |
| 0.0      | 421             | 1         | 421.0         | 421.0         | 421         | 421         | 0.0         | cuLibraryGetKernel     |

[6/8] Executing 'cuda\_gpu\_kern\_sum' stats report

| Time (%) | Total Time (ns) | Instances | Avg (ns)    | Med (ns)    | Min (ns)  | Max (ns)  | StdDev (ns) | Name                          |
|----------|-----------------|-----------|-------------|-------------|-----------|-----------|-------------|-------------------------------|
| 100.0    | 12,333,088      | 10        | 1,233,308.8 | 1,227,149.0 | 1,133,772 | 1,370,670 | 81,101.7    | bodyForce(Body *, float, int) |

[7/8] Executing 'cuda\_gpu\_mem\_time\_sum' stats report

| Time (%) | Total Time (ns) | Count | Avg (ns) | Med (ns) | Min (ns) | Max (ns) | StdDev (ns) | Operation                            |
|----------|-----------------|-------|----------|----------|----------|----------|-------------|--------------------------------------|
| 60.9     | 216,526         | 36    | 6,014.6  | 6,415.5  | 2,207    | 9,343    | 2,916.7     | [CUDA memcpy Unified Host-to-Device] |
| 39.1     | 139,150         | 40    | 3,478.8  | 3,535.5  | 1,375    | 5,824    | 2,062.4     | [CUDA memcpy Unified Device-to-Host] |

[8/8] Executing 'cuda\_gpu\_mem\_size\_sum' stats report

| Total (MB) | Count | Avg (MB) | Med (MB) | Min (MB) | Max (MB) | StdDev (MB) | Operation                            |
|------------|-------|----------|----------|----------|----------|-------------|--------------------------------------|
| 1.311      | 40    | 0.033    | 0.033    | 0.004    | 0.061    | 0.029       | [CUDA memcpy Unified Device-to-Host] |
| 1.180      | 36    | 0.033    | 0.033    | 0.004    | 0.061    | 0.024       | [CUDA memcpy Unified Host-to-Device] |

## average execution time for v6\_2.cu (65536)

1.7948 sec

## nsys profile for [v7.cu](#) (4096)

[5/8] Executing 'cuda\_api\_sum' stats report

| Time (%) | Total Time (ns) | Num Calls | Avg (ns)      | Med (ns)      | Min (ns)    | Max (ns)    | StdDev (ns) | Name                   |
|----------|-----------------|-----------|---------------|---------------|-------------|-------------|-------------|------------------------|
| 92.4     | 166,732,272     | 1         | 166,732,272.0 | 166,732,272.0 | 166,732,272 | 166,732,272 | 0.0         | cudaMallocManaged      |
| 7.1      | 12,777,125      | 10        | 1,277,712.5   | 1,279,831.5   | 1,126,641   | 1,350,414   | 60,795.9    | cudaDeviceSynchronize  |
| 0.2      | 446,368         | 10        | 44,636.8      | 5,895.0       | 5,188       | 382,433     | 118,747.5   | cudaLaunchKernel       |
| 0.2      | 381,505         | 1         | 381,505.0     | 381,505.0     | 381,505     | 381,505     | 0.0         | cuLibraryLoadData      |
| 0.1      | 147,468         | 1         | 147,468.0     | 147,468.0     | 147,468     | 147,468     | 0.0         | cudaFree               |
| 0.0      | 3,251           | 10        | 325.1         | 226.5         | 154         | 977         | 259.5       | cuKernelGetName        |
| 0.0      | 1,413           | 1         | 1,413.0       | 1,413.0       | 1,413       | 1,413       | 0.0         | cuModuleGetLoadingMode |
| 0.0      | 463             | 1         | 463.0         | 463.0         | 463         | 463         | 0.0         | cuLibraryGetKernel     |

[6/8] Executing 'cuda\_gpu\_kern\_sum' stats report

| Time (%) | Total Time (ns) | Instances | Avg (ns)    | Med (ns)    | Min (ns)  | Max (ns)  | StdDev (ns) | Name                          |
|----------|-----------------|-----------|-------------|-------------|-----------|-----------|-------------|-------------------------------|
| 100.0    | 10,806,406      | 10        | 1,080,640.6 | 1,072,173.5 | 1,058,381 | 1,132,110 | 23,359.9    | bodyForce(Body *, float, int) |

[7/8] Executing 'cuda\_gpu\_mem\_time\_sum' stats report

| Time (%) | Total Time (ns) | Count | Avg (ns) | Med (ns) | Min (ns) | Max (ns) | StdDev (ns) | Operation                            |
|----------|-----------------|-------|----------|----------|----------|----------|-------------|--------------------------------------|
| 69.5     | 317,556         | 90    | 3,528.4  | 3,535.0  | 2,271    | 8,672    | 957.2       | [CUDA memcpy Unified Host-to-Device] |
| 30.5     | 139,405         | 40    | 3,485.1  | 3,471.5  | 1,407    | 5,857    | 2,084.8     | [CUDA memcpy Unified Device-to-Host] |

[8/8] Executing 'cuda\_gpu\_mem\_size\_sum' stats report

| Total (MB) | Count | Avg (MB) | Med (MB) | Min (MB) | Max (MB) | StdDev (MB) | Operation                            |
|------------|-------|----------|----------|----------|----------|-------------|--------------------------------------|
| 1.311      | 40    | 0.033    | 0.033    | 0.004    | 0.061    | 0.029       | [CUDA memcpy Unified Device-to-Host] |
| 1.180      | 90    | 0.013    | 0.012    | 0.004    | 0.033    | 0.008       | [CUDA memcpy Unified Host-to-Device] |

average execution time for v7\_2.cu (65536)  
0.6964 sec

## nsys profile for [v8.cu](#) (4096)

[5/8] Executing 'cuda\_api\_sum' stats report

| Time (%) | Total Time (ns) | Num Calls | Avg (ns)      | Med (ns)      | Min (ns)    | Max (ns)    | StdDev (ns) | Name                   |
|----------|-----------------|-----------|---------------|---------------|-------------|-------------|-------------|------------------------|
| 94.2     | 163,693,271     | 1         | 163,693,271.0 | 163,693,271.0 | 163,693,271 | 163,693,271 | 0.0         | cudaMallocManaged      |
| 4.9      | 8,589,739       | 10        | 858,973.9     | 811,875.0     | 781,295     | 1,174,293   | 117,669.0   | cudaDeviceSynchronize  |
| 0.4      | 638,827         | 10        | 63,882.7      | 25,069.0      | 15,830      | 360,203     | 105,621.7   | cudaLaunchKernel       |
| 0.3      | 463,559         | 1         | 463,559.0     | 463,559.0     | 463,559     | 463,559     | 0.0         | cuLibraryLoadData      |
| 0.2      | 301,793         | 1         | 301,793.0     | 301,793.0     | 301,793     | 301,793     | 0.0         | cudaFree               |
| 0.0      | 53,497          | 10        | 5,349.7       | 880.0         | 505         | 45,589      | 14,141.4    | cuKernelGetName        |
| 0.0      | 1,583           | 1         | 1,583.0       | 1,583.0       | 1,583       | 1,583       | 0.0         | cuModuleGetLoadingMode |
| 0.0      | 399             | 1         | 399.0         | 399.0         | 399         | 399         | 0.0         | cuLibraryGetKernel     |

[6/8] Executing 'cuda\_gpu\_kern\_sum' stats report

| Time (%) | Total Time (ns) | Instances | Avg (ns)  | Med (ns)  | Min (ns) | Max (ns) | StdDev (ns) | Name                          |
|----------|-----------------|-----------|-----------|-----------|----------|----------|-------------|-------------------------------|
| 100.0    | 6,424,607       | 10        | 642,460.7 | 590,345.0 | 572,105  | 878,797  | 104,458.6   | bodyForce(Body *, float, int) |

[7/8] Executing 'cuda\_gpu\_mem\_time\_sum' stats report

| Time (%) | Total Time (ns) | Count | Avg (ns) | Med (ns) | Min (ns) | Max (ns) | StdDev (ns) | Operation                            |
|----------|-----------------|-------|----------|----------|----------|----------|-------------|--------------------------------------|
| 56.9     | 189,210         | 18    | 10,511.7 | 10,287.5 | 4,352    | 17,599   | 6,137.6     | [CUDA memcpy Unified Host-to-Device] |
| 43.1     | 143,373         | 40    | 3,584.3  | 3,439.5  | 1,407    | 9,696    | 2,292.2     | [CUDA memcpy Unified Device-to-Host] |

[8/8] Executing 'cuda\_gpu\_mem\_size\_sum' stats report

| Total (MB) | Count | Avg (MB) | Med (MB) | Min (MB) | Max (MB) | StdDev (MB) | Operation                            |
|------------|-------|----------|----------|----------|----------|-------------|--------------------------------------|
| 1.311      | 40    | 0.033    | 0.033    | 0.004    | 0.061    | 0.029       | [CUDA memcpy Unified Device-to-Host] |
| 1.180      | 18    | 0.066    | 0.066    | 0.033    | 0.098    | 0.034       | [CUDA memcpy Unified Host-to-Device] |

average execution time for v8\_2.cu (65536)  
0.9194 sec

## nsys profile for [v9.cu](#) (4096)

[5/8] Executing 'cuda\_api\_sum' stats report

| Time (%) | Total Time (ns) | Num Calls | Avg (ns)      | Med (ns)      | Min (ns)    | Max (ns)    | StdDev (ns) | Name                   |
|----------|-----------------|-----------|---------------|---------------|-------------|-------------|-------------|------------------------|
| 94.6     | 196,921,298     | 1         | 196,921,298.0 | 196,921,298.0 | 196,921,298 | 196,921,298 | 0.0         | cudaMallocManaged      |
| 4.7      | 9,816,860       | 10        | 981,686.0     | 963,097.5     | 783,503     | 1,161,959   | 103,461.8   | cudaDeviceSynchronize  |
| 0.4      | 792,052         | 10        | 79,205.2      | 35,623.5      | 10,048      | 442,169     | 130,520.2   | cudaLaunchKernel       |
| 0.2      | 372,048         | 1         | 372,048.0     | 372,048.0     | 372,048     | 372,048     | 0.0         | cuLibraryLoadData      |
| 0.1      | 177,862         | 1         | 177,862.0     | 177,862.0     | 177,862     | 177,862     | 0.0         | cudaFree               |
| 0.0      | 12,997          | 10        | 1,299.7       | 1,573.0       | 217         | 1,814       | 615.7       | cuKernelGetName        |
| 0.0      | 1,491           | 1         | 1,491.0       | 1,491.0       | 1,491       | 1,491       | 0.0         | cuModuleGetLoadingMode |
| 0.0      | 386             | 1         | 386.0         | 386.0         | 386         | 386         | 0.0         | cuLibraryGetKernel     |

[6/8] Executing 'cuda\_gpu\_kern\_sum' stats report

| Time (%) | Total Time (ns) | Instances | Avg (ns)  | Med (ns)  | Min (ns) | Max (ns) | StdDev (ns) | Name                          |
|----------|-----------------|-----------|-----------|-----------|----------|----------|-------------|-------------------------------|
| 100.0    | 7,910,909       | 10        | 791,090.9 | 760,604.0 | 714,699  | 951,855  | 78,764.7    | bodyForce(Body *, float, int) |

[7/8] Executing 'cuda\_gpu\_mem\_time\_sum' stats report

| Time (%) | Total Time (ns) | Count | Avg (ns) | Med (ns) | Min (ns) | Max (ns) | StdDev (ns) | Operation                            |
|----------|-----------------|-------|----------|----------|----------|----------|-------------|--------------------------------------|
| 58.9     | 225,692         | 18    | 12,538.4 | 14,575.5 | 4,385    | 42,592   | 9,535.8     | [CUDA memcpy Unified Host-to-Device] |
| 41.1     | 157,581         | 40    | 3,939.5  | 3,583.5  | 1,407    | 13,249   | 2,942.0     | [CUDA memcpy Unified Device-to-Host] |

[8/8] Executing 'cuda\_gpu\_mem\_size\_sum' stats report

| Total (MB) | Count | Avg (MB) | Med (MB) | Min (MB) | Max (MB) | StdDev (MB) | Operation                            |
|------------|-------|----------|----------|----------|----------|-------------|--------------------------------------|
| 1.311      | 40    | 0.033    | 0.033    | 0.004    | 0.061    | 0.029       | [CUDA memcpy Unified Device-to-Host] |
| 1.180      | 18    | 0.066    | 0.066    | 0.033    | 0.098    | 0.034       | [CUDA memcpy Unified Host-to-Device] |

average execution time for v9\_2.cu (65536)

0.4850 sec

## nsys profile for [v10.cu](#) (4096)

[5/8] Executing 'cuda\_api\_sum' stats report

| Time (%) | Total Time (ns) | Num Calls | Avg (ns)      | Med (ns)      | Min (ns)    | Max (ns)    | StdDev (ns) | Name                   |
|----------|-----------------|-----------|---------------|---------------|-------------|-------------|-------------|------------------------|
| 96.1     | 140,500,792     | 1         | 140,500,792.0 | 140,500,792.0 | 140,500,792 | 140,500,792 | 0.0         | cudaHostAlloc          |
| 2.8      | 4,114,879       | 10        | 411,487.9     | 413,892.0     | 389,006     | 414,992     | 7,925.2     | cudaDeviceSynchronize  |
| 0.3      | 441,260         | 1         | 441,260.0     | 441,260.0     | 441,260     | 441,260     | 0.0         | cudaFreeHost           |
| 0.3      | 415,446         | 20        | 20,772.3      | 3,764.0       | 3,524       | 307,335     | 67,782.0    | cudaLaunchKernel       |
| 0.3      | 384,093         | 1         | 384,093.0     | 384,093.0     | 384,093     | 384,093     | 0.0         | cuLibraryLoadData      |
| 0.1      | 123,370         | 1         | 123,370.0     | 123,370.0     | 123,370     | 123,370     | 0.0         | cudaFree               |
| 0.1      | 103,014         | 1         | 103,014.0     | 103,014.0     | 103,014     | 103,014     | 0.0         | cudaMalloc             |
| 0.1      | 75,219          | 2         | 37,609.5      | 37,609.5      | 24,197      | 51,022      | 18,968.1    | cudaMemcpy             |
| 0.0      | 2,796           | 20        | 139.8         | 110.0         | 103         | 473         | 82.6        | cuKernelGetName        |
| 0.0      | 1,091           | 1         | 1,091.0       | 1,091.0       | 1,091       | 1,091       | 0.0         | cuModuleGetLoadingMode |
| 0.0      | 770             | 2         | 385.0         | 385.0         | 303         | 467         | 116.0       | cuLibraryGetKernel     |

[6/8] Executing 'cuda\_gpu\_kern\_sum' stats report

| Time (%) | Total Time (ns) | Instances | Avg (ns)  | Med (ns)  | Min (ns) | Max (ns) | StdDev (ns) | Name                                   |
|----------|-----------------|-----------|-----------|-----------|----------|----------|-------------|----------------------------------------|
| 99.5     | 4,125,791       | 10        | 412,579.1 | 412,566.5 | 411,750  | 413,638  | 622.9       | bodyForce(Body *, float, int)          |
| 0.5      | 22,017          | 10        | 2,201.7   | 2,176.0   | 2,080    | 2,528    | 121.4       | integratePositions(Body *, float, int) |

[7/8] Executing 'cuda\_gpu\_mem\_time\_sum' stats report

| Time (%) | Total Time (ns) | Count | Avg (ns) | Med (ns) | Min (ns) | Max (ns) | StdDev (ns) | Operation                    |
|----------|-----------------|-------|----------|----------|----------|----------|-------------|------------------------------|
| 68.6     | 18,752          | 1     | 18,752.0 | 18,752.0 | 18,752   | 18,752   | 0.0         | [CUDA memcpy Host-to-Device] |
| 31.4     | 8,576           | 1     | 8,576.0  | 8,576.0  | 8,576    | 8,576    | 0.0         | [CUDA memcpy Device-to-Host] |

[8/8] Executing 'cuda\_gpu\_mem\_size\_sum' stats report

| Total (MB) | Count | Avg (MB) | Med (MB) | Min (MB) | Max (MB) | StdDev (MB) | Operation                    |
|------------|-------|----------|----------|----------|----------|-------------|------------------------------|
| 0.098      | 1     | 0.098    | 0.098    | 0.098    | 0.098    | 0.000       | [CUDA memcpy Device-to-Host] |
| 0.098      | 1     | 0.098    | 0.098    | 0.098    | 0.098    | 0.000       | [CUDA memcpy Host-to-Device] |

average execution time for v10\_2.cu (65536)

0.4619 sec